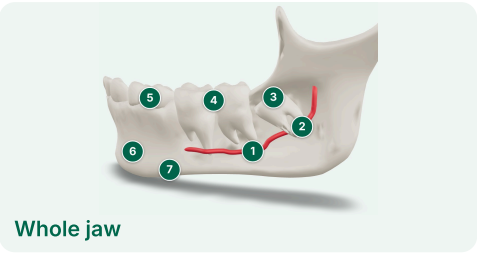
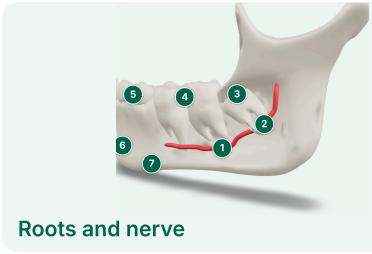
Impacted lower wisdom tooth and the nerve near it

NAGOYA UNIVERSITY

Every person is different
The shape of the tooth, where the nerve is, and the risks can be very different from person to person. So the facts here are general. They are not advice for one person. Before any treatment, the dentist or surgeon will talk with each person in detail about the benefits, risks and choices.



Whole jaw



Roots and nerve


From below

Jaw bone and teeth Nerve, artery and vein (in their canal)

Parts of the model

1 Jaw nerve (inferior alveolar nerve)
- Gives feeling to the lower teeth, gums, lip and chin on this side.
- Runs in a tube in the bone with the inferior alveolar artery and vein. These can also be hurt.
- If the nerve is hurt, the area can feel numb, have pins and needles, feel strange or hurt. This can be short-term or long-term.
- In older adults, damage is more likely, and less likely to heal fully.

2 Roots of the wisdom tooth
- The roots wrap around the jaw nerve.
- So taking the tooth out can hurt the nerve.

3 Wisdom tooth (third molar)
- Impacted: stuck under the gum and bone.
- Unerupted: it has not come through the gum.

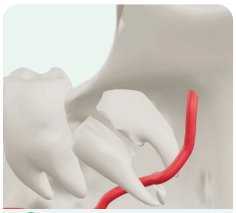
4 First and second molars
- Big back teeth used to chew.
- The wisdom tooth sits behind them.

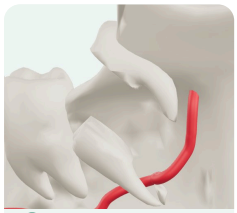
5 First and second premolars
- The teeth in front of the molars.
- They also help to chew.

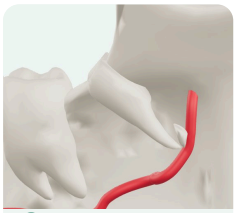
6 Chin nerve (mental nerve)
- A branch of the jaw nerve. It comes out of the bone through a small hole (the mental foramen).
- Gives feeling to the lower lip and chin.
- If the jaw nerve is hurt further back, the lip and chin can feel different too.

7 Jaw bone (mandible)
- The jaw nerve, artery and vein run in a tube inside this bone.

How the tooth can come out in pieces


1 The tooth is cut into 2 pieces


2 The back (distal) piece comes out


3 Then the front (mesial) piece comes out

What is the problem?

The wisdom tooth is the last tooth at the back. Its other name is the **third molar**. An **impacted lower wisdom tooth** is stuck under the gum and bone. It has not come through the gum (it is **unerupted**). It cannot grow in the right way.

The nerve

A nerve runs in a tube of bone inside the lower jaw. Here it is called the **jaw nerve**. Its medical name is the **inferior alveolar nerve**. Two blood vessels run with it in the same tube. They are the **inferior alveolar artery and vein**.[1]

The jaw nerve gives feeling to the lower teeth and gums on the same side.[2] Near the front of the jaw, a branch of it comes out of the bone through a small hole. This branch is the **chin nerve** (**mental nerve**). It gives feeling to the lower lip and chin.[3]

The tongue gets its feeling from a different nerve, the **lingual nerve**. It can also be hurt when a wisdom tooth is taken out.[4] Lasting damage to it is rare. It happens to about 1 to 3 in 1,000 people.[5] The tongue nerve is not shown in this model.

In this model, **the roots of the wisdom tooth wrap around the nerve and artery**.

Why taking the tooth out can hurt the nerve

To take the tooth out, the roots must come out of the bone. The roots curve around the nerve. So the nerve can be pressed, stretched or cut as the roots come out. The artery and vein can also be hurt and bleed.[1]

If the jaw nerve is hurt, the chin nerve may also stop working well. This is because it is a branch of the jaw nerve.[3]

What nerve damage feels like

The lower lip, chin, teeth or gums on that side may:[2]
- feel numb
- tingle, like pins and needles
- feel different, or strange
- hurt. Even a light touch may hurt.[4]

The lip and face can still move as normal.[2]

Short-term: the feeling comes back. This most often takes weeks to months.[2]

Long-term: the feeling does not fully come back.[2]

Age: the risk of nerve damage is higher in people over about 24 years old.[2] In older people, the feeling is also less likely to come back fully.[6] One small study followed people for 5 years. Everyone under 30 got full feeling back. Only 3 of 8 people over 30 did.[7] Studies do not agree on one exact age.

How often does it happen?

Some wisdom teeth have roots close to the nerve. A review of 16 studies looked at these teeth. It found:[8]

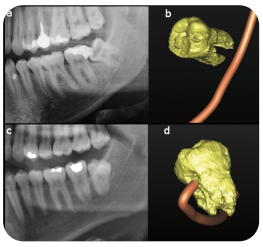
About 7 in 100
had nerve damage after the whole tooth was taken out

About 1 in 100
had nerve damage that did not go away

Why a 3D scan can help

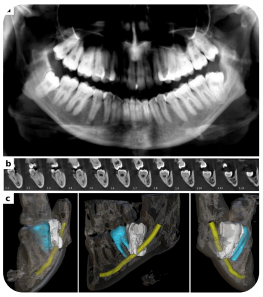
The usual dental X-ray of the whole mouth is called an **OPG**. OPG is short for **orthopantomogram**. It is also called a panoramic X-ray. It is a flat picture. On a flat picture, the roots can look like they touch the nerve, even when they do not.[9] Some signs on an OPG can warn that the roots are close to the nerve.[10]

A **CBCT** scan is a 3D X-ray. CBCT is short for **cone beam computed tomography**. It can show if the roots really touch the nerve. It also shows which side of the roots the nerve is on.[9] Studies show it does this better than an OPG.[11]



The two OPGs (a and c) look much the same. The CBCT images show they are not. Top (b): the tooth does not touch the nerve. Bottom (d): the roots wrap around the nerve. In b and d, the tooth is yellow and the nerve is orange.[9] Picture: Rodriguez y Baena R, et al. J Clin Exp Dent. 2017, Figure 1. Licence: CC BY 2.5. Not changed.

The next pictures show what each scan looks like. They are of one person's lower wisdom tooth. The nerve runs between its roots.[12]



a: an OPG. It shows the whole mouth in one flat picture. **b:** a CBCT scan. It shows the jaw as many thin slices, like slices of bread. The small dark circle in the bone is the tube that holds the nerve. **c:** a computer can turn the CBCT slices into a 3D picture. The wisdom tooth is white. The nerve is yellow.[12] Pictures: Marquez Rocco V, et al. Clin Case Rep. 2026, Figures 1, 2 (top row) and 3. Licence: CC BY 4.0. Cropped, put together, and letters a to c added.

A CBCT scan uses more X-ray than an OPG. So it is used when an OPG shows the roots may be close to the nerve.[9]

Risks of taking the tooth out

After the tooth is taken out, it is common to have pain, swelling and a stiff jaw that is hard to open. These most often last only a short time.[13]

Less often, the hole where the tooth was can get infected. Or the blood clot in the hole can come out. This leaves a painful "dry socket".[13] The nerves can also be hurt, as shown above.[13]

Risks of leaving the tooth in

A wisdom tooth that stays in can cause problems later. These include:[14]

- infection of the gum over the tooth
- gum disease or decay in the tooth in front
- harm to the roots of the tooth in front
- a **cyst** (a sac of fluid) around the tooth

In one review, about 4 in 100 impacted wisdom teeth that were taken out had a cyst.[15] So a tooth that stays in needs regular check-ups.[14]

The dentist or surgeon will talk in detail about the risks and the best choice for each person.

Choices your dentist/surgeon might offer

- **Leave the tooth in and check it often**, if it is not causing problems.
- **Coronectomy:** take out only the top of the tooth. Leave the roots around the nerve. In the same review, about 1 in 200 people had nerve damage after this.[8] But the roots can move. Once in a while they need to come out later.
- **Take out the whole tooth.** This is often a **surgical extraction**. A small cut is made in the gum. If bone is in the way of the tooth, some of it may be removed. In harder cases, the tooth may be cut into pieces first. For example, it can be cut into a front (**mesial**) piece and a back (**distal**) piece. Then each piece comes out on its own.[16]

Before any treatment, the dentist or surgeon will explain the benefits, risks and choices in much more detail.

Questions you could ask your dentist/surgeon

- How close are the roots to the nerve?
- What is the risk of nerve damage in this case?
- Is a CBCT scan needed?
- Is a coronectomy a good choice in this case?
- What happens if the tooth is left in?

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